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The Editors  
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Dear Editors,

On behalf of my coauthors, I am pleased to submit the manuscript “Representation as a Computational Resource in Quantum Compilation: Approximate Recoverability Tradeoffs and Model-Specific Fault-Tolerant Consequences” for consideration in *PRX Quantum*.

Fault-tolerant compilation deliberately destroys structure. Time-dependent and variational schedules split one accumulated phase across many rounds; gate synthesis dissolves an angle into a Clifford+ $T$  word in which no individual letter carries the angle; randomized compiling twirls each rotation into sign-randomized fragments. Each transformation is correct and desirable for its own reason, and each scatters a quantity that was once written in one place across an instruction stream. The literature on phase-polynomial, Pauli-gadget, and ZX methods establishes that a sufficiently capable pass can sometimes reassemble such a quantity. It does not establish what that reassembly must cost. This manuscript answers that question.

The central result is a lower bound with two admissible payment channels. We formalize a downstream compiler—one positioned after randomization and lowering, as a cloud transpiler or a control-stack stream optimizer is—and charge every restart channel in literal bits: control, store, live window, parameters, and intermediate representation. For a commuting phase layer with  $m$  generators compiled to projective accuracy  $\epsilon$ , any such compiler making  $p$  forward passes and failing with probability at most  $\delta$  obeys

$$\overline{A}_p + (2p - 1)S \geq (1 - \delta)m \log_2 K_\epsilon - h_2(\delta),$$

where  $\overline{A}_p$  is committed output and  $S$  the serialized crossing state. The scattered information must be bought back either in bits of compiler state or in magic states, and there is no third option. Because a memory-capped block compiler attains the envelope within a constant factor, the frontier is tight rather than merely a floor. Charging the intermediate representation explicitly is what makes the statement non-evadable: a ZX graph, a phase-polynomial table, a cache, or a spilled update log is an accounted channel rather than a free side door.

A lower bound of this kind would be of limited interest if its hypotheses were met only by a constructed secret sharing. They are not. Time-dependent scheduling supplies the additive decomposition, randomized compiling supplies the sign symmetry, and the Ross–Selinger synthesis grid supplies the modulus, so the hypotheses are satisfied by streams that a fault-tolerant stack emits anyway. Under a disclosed frame-tracked presentation the crossing state is the Pauli frame itself, which yields a tension we expect will interest the noise-tailoring community: the randomization that averages coherent noise is the same randomization that forces a re-aggregating compiler to carry  $\Omega(m \log(1/\epsilon))$  bits. The boundaries of the claim are stated explicitly; the

non-commuting interleaving of a generic Trotter step lies outside them and is not asserted.

A refinement shows that the two currencies are not worth the same. Once a committed record is executed rather than buffered, the committed channel additionally pays  $\log_2 \binom{m}{k}$ —the entropy of the compiler’s own decision about which  $k$  coordinates to commit early—while a deferring compiler is handed that decision for free by the stream that follows. The toll is independent of  $m$  and vanishes exactly when the commitment schedule is public. It was measured on an instrumented device under a preregistration frozen before the run, and matches the predicted entropy to within 0.17 bits at the worst cell.

The paper does not stop at the information-theoretic statement. Instrumented reference compilers expose the state, IR, and commitment coordinates directly at the bit level, and one fixed total error budget is propagated through actual `staq grid_synth` Clifford+ $T$  synthesis and executable surface-code scheduling. Under the disclosed model, the same unitary compiled from a dispersed rather than an aggregate representation costs 385–1,753 times the logical  $T$  states and 538–5,573 times the spacetime volume, with no reversal anywhere in the scanned error and factory grid. The representational gap becomes a fault-tolerant one.

We believe the work suits *PRX Quantum* because it sits across communities that rarely share a paper—compiler and IR design, randomized compiling and noise tailoring, fault-tolerant resource estimation, and classical communication complexity—and because its conclusion is a design rule with a proof attached: randomize and lower late. It also identifies representation as a controlled variable that resource-estimation studies should report rather than leave implicit. The stance toward existing tools is explanatory rather than competitive. TKET’s `PauliSimp` and PyZX’s `full_reduce` already implement the semantic-first principle on their global intermediate representations; the contribution here is the bound that says what any such tool must pay to do so on a dispersed stream. Their recovery is measured and credited—external TKET uses 0.443–0.536 times the reference’s  $T$  states on the same witness—and no headline ranks our implementation above them. In the same spirit, two preregistered attempts to carry the commitment toll onto third-party compilers failed, and are reported as failures rather than repaired.

Every headline number is generated from a normalized frozen-data registry with a numeric provenance map and a one-command entry point. Correctness is gated by an independent symbolic or dense certificate, status outcomes are never converted into numerical wins, and a clean-room runner reconstructs the theorem, benchmark, and resource-estimation subsets in an independent directory.

This manuscript is original and previously unpublished. It has not been posted as a preprint, and it is not under consideration at any other journal. We declare no competing interests.

Thank you for your consideration. We would be glad to answer any questions.

Sincerely,

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*on behalf of all authors*